

BOULOGNE, France

WMO: 070020

Lat: 50.7325N

Lon: 1.5997E

Elev: 74

StdP: 100.44

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-1.9	-8.3	1.9	-1.7	-6.3	2.2	-0.8	17.5	9.0	15.4	8.5	4.4	70	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	26.3	18.9	24.0	17.9	22.2	17.0	19.7	24.7	18.6	22.5	17.8	20.9	3.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.9	21.3	17.1	12.3	20.1	16.5	11.8	19.4	56.5	25.0	53.1	22.5	50.3	21.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.6	11.3	-5.0	30.6	2.7	2.6	-6.9	32.5	-8.5	34.0	-10.0	35.5	-11.9	37.3		
			-6.0	21.9	2.5	0.9	-7.8	22.6	-9.3	23.1	-10.7	23.7	-12.6	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	5.1	5.3	7.2	9.8	12.5	15.2	17.2	17.8	15.9	12.8	8.9	6.1
	DBStd	5.37	3.54	3.20	2.88	3.10	3.04	2.64	2.51	2.35	2.20	2.78	2.90	3.55
	HDD10.0	612	153	133	93	39	7	0	0	0	0	8	53	126
	HDD18.3	2666	411	364	344	257	183	102	53	37	80	172	284	380
	CDD10.0	1044	1	1	7	33	86	156	224	243	176	96	19	4
	CDD18.3	58	0	0	0	1	3	8	18	22	6	1	0	0
	CDH23.3	268	0	0	0	1	11	44	93	103	16	1	0	0
	CDH26.7	57	0	0	0	0	1	9	22	22	2	0	0	0
Wind	WSAvg	5.5	6.4	6.1	5.6	4.9	5.0	4.7	5.0	5.0	5.1	5.8	6.0	6.5
Precipitation	PrecAvg	857	78	59	49	52	58	55	59	66	72	97	111	101
	PrecMax	1158	146	155	153	147	122	132	134	135	161	242	264	205
	PrecMin	572	10	15	6	2	6	23	8	11	2	23	32	24
	PrecStd	153	36	35	31	35	29	29	30	39	39	51	55	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.8	17.2	21.7	24.7	27.8	29.2	29.6	25.5	21.8	16.1	13.1
		MCWB	10.1	9.1	11.1	13.8	16.6	19.4	19.7	20.4	18.4	16.6	13.8	11.4
	2%	DB	10.9	11.2	14.3	18.8	22.1	24.0	25.8	26.2	22.3	18.9	14.5	12.2
		MCWB	9.7	8.5	10.3	12.6	16.0	17.8	18.5	19.1	17.3	15.7	12.9	10.9
	5%	DB	10.2	10.0	12.4	16.5	19.5	21.5	23.3	23.7	20.3	17.3	13.7	11.5
		MCWB	9.0	8.6	9.4	11.5	14.4	16.6	17.8	18.2	16.3	15.0	12.2	10.3
	10%	DB	9.4	9.2	10.7	14.2	17.4	19.3	21.1	21.4	18.8	16.3	12.9	10.7
		MCWB	8.2	8.0	8.5	10.5	13.6	15.4	16.9	17.2	15.7	14.3	11.3	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.2	12.2	14.6	18.4	20.1	21.1	21.4	19.3	17.2	14.6	12.1
		MCDB	11.3	11.5	15.1	19.6	22.7	26.3	27.9	27.6	23.2	19.8	15.4	12.6
	2%	WB	10.0	9.4	10.9	13.2	16.4	18.5	19.5	19.8	18.1	16.2	13.4	11.2
		MCDB	10.7	10.5	13.4	17.8	20.5	23.1	24.4	24.4	21.0	18.2	14.3	11.9
	5%	WB	9.2	8.8	9.8	12.0	14.9	17.0	18.2	18.7	17.2	15.4	12.5	10.5
		MCDB	9.9	9.7	11.7	15.5	18.5	20.8	21.9	22.4	19.2	17.0	13.5	11.3
	10%	WB	8.4	8.1	9.0	10.9	13.7	15.8	17.2	17.8	16.5	14.7	11.5	9.6
		MCDB	9.3	9.1	10.3	13.7	16.6	18.7	20.1	20.6	18.2	16.1	12.7	10.5
Mean Daily Temperature Range	5% DB	MDBR	3.5	3.6	4.3	5.3	5.2	5.2	4.9	4.8	4.8	4.3	3.6	3.5
		MCDBR	3.6	4.3	7.0	8.4	8.8	9.3	9.2	8.7	7.1	5.4	4.0	3.3
		MCWBR	3.7	3.4	4.5	4.8	5.1	5.4	4.7	4.3	3.9	3.7	3.7	3.4
	5% WB	MCDBR	3.4	3.6	5.7	7.6	8.0	8.4	7.9	7.1	5.5	4.8	3.6	3.3
		MCWBR	3.7	3.2	4.1	4.6	5.0	5.3	4.5	4.0	3.6	3.9	3.8	3.6
Clear-Sky Solar Irradiance	taub	0.329	0.346	0.383	0.402	0.410	0.410	0.407	0.400	0.379	0.364	0.338	0.322	
	taud	2.394	2.385	2.284	2.255	2.288	2.312	2.324	2.367	2.408	2.431	2.417	2.380	
	Ebn at Noon	693	780	810	837	847	849	844	832	809	750	675	638	
	Edh at Noon	64	82	108	124	125	123	120	110	95	78	62	56	
All-Sky Solar Radiation	RadAvg	0.83	1.54	2.75	4.42	5.24	5.80	5.56	4.65	3.46	1.96	0.96	0.64	
	RadStd	0.07	0.22	0.33	0.38	0.37	0.48	0.45	0.32	0.22	0.18	0.10	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page